

Kiwi.com Online Ticket Booking Application - Testing Checklist

1. FUNCTIONAL TESTING

1.1 Flight Search & Discovery Features

A. Basic Flight Search Functionality

Testing Method: Black Box Testing

Testing Level: System Testing

Smoke Testing:

- Basic search with origin/destination works
- Date selection functionality works
- Search results display properly
- One-way vs round-trip toggle works

Critical Path Testing:

- Search with popular routes (London-Paris, NYC-LA)
- Return flight booking flow
- Price sorting and filtering
- Mobile search functionality

Extended Testing:

- Positive Tests:
 - Valid airport codes (IATA/ICAO)
 - Future date selections
 - Multi-city search combinations
 - Special character handling in city names
- Negative Tests:
 - Invalid airport codes
 - Past dates selection
 - Empty search fields

- Boundary testing: 150,000 characters in search field
- Special characters and SQL injection attempts

B. "Anywhere" & "Anytime" Features

Testing Method: Black Box Testing

Testing Level: System & Integration Testing

- "Anywhere" destination search from specific origin
- "Anytime" flexible date functionality
- Map integration with search results
- Results accuracy and data validation

1.2 Booking Process Flow

A. Flight Selection & Review

Testing Method: Black Box Testing

Testing Level: System Testing

- Flight selection and price accuracy
- Seat selection integration
- Baggage options functionality
- Insurance add-ons
- Multi-airline booking coordination

B. Passenger Information Management

Testing Method: Black Box Testing

Testing Level: System & Unit Testing

Positive Testing:

- Valid passenger details acceptance
- Multiple passenger booking
- Auto-fill from saved profiles
- Contact information validation

Negative Testing:

- Invalid passport formats
- Missing required fields
- Boundary testing on name lengths
- Special characters in names
- Invalid email formats

C. Payment Processing

Testing Method: Grey Box Testing (partial payment gateway knowledge)

Testing Level: Integration & System Testing

- Multiple payment methods validation
- Currency conversion accuracy
- Payment security protocols
- Failed payment handling
- Refund processing workflows

2. NON-FUNCTIONAL TESTING

2.1 Performance Testing

A. Load Testing

Testing Level: System Testing

- Gradual increase in the number of users (every 20 seconds)
- Normal business hours traffic simulation
- Peak booking periods (holiday seasons)
- Search response time under load
- Database query performance

B. Stress Testing

Testing Level: System Testing

- Simultaneous user login scenarios
- Limited memory/server access conditions

- Critical breaking points identification
- Recovery mechanisms validation

C. Spike Testing

Testing Level: System Testing

- Flash sale traffic spikes
- Promotional campaign traffic bursts
- Return to steady state validation

D. Volume Testing

Testing Level: System & Integration Testing

- Large flight databases
- Historical booking data impact
- Search performance with millions of routes
- Data archiving and retrieval

E. Endurance Testing

Testing Level: System Testing

- 8-hour continuous operation
- 24-hour reliability testing
- Memory leak detection
- Resource usage monitoring

F. Scalability Testing

Testing Level: System Testing

- Horizontal scaling validation(Adding more servers to distribute the load)
- Load balancer effectiveness
- CDN optimization(Content Delivery Network)

2.2 Security Testing

Testing Method: Grey Box Testing

Testing Level: System Testing

- Payment data encryption (PCI DSS compliance)
- User authentication security
- Session management and timeouts
- SQL injection prevention
- HTTPS/SSL implementation
- Personal data protection (General Data Protection Regulation)

2.3 Usability Testing

Testing Method: Black Box Testing

Testing Level: System Testing

- Two-click purchase capability
- Intuitive navigation flows
- Error message clarity
- Mobile interface usability
- Booking modification ease

2.4 Compatibility Testing

Testing Method: Black Box Testing

Testing Level: System Testing

- Cross-browser functionality (Chrome, Safari, Edge, Firefox)
- Mobile OS compatibility (iOS 14+, Android 10+)
- Desktop OS compatibility (Windows 10+, macOS)
- Different screen resolutions
- Touch vs mouse interactions

2.5 Accessibility Testing

Testing Method: Black Box Testing

Testing Level: System Testing

- Screen reader compatibility
- High contrast mode functionality
- Keyboard navigation support
- Text scaling capabilities
- Color blindness considerations
- Voice access integration

2.6 Internationalization & Localization Testing

Testing Method: Black Box Testing

Testing Level: System Testing

Internationalization Testing:

- Multi-language interface support
- Currency conversion accuracy
- Date/time format adaptations
- Right-to-left language support

Localization Testing:

- Translation quality and completeness
- Cultural adaptations
- Local payment methods
- Regional compliance requirements

3. STRUCTURAL TESTING

3.1 Code Structure Analysis

Testing Method: White Box Testing

Testing Level: Unit & Component Testing

Performed By: Developers

- Statement coverage for booking calculations
- Decision coverage for pricing logic
- Path coverage for search algorithms
- Data flow testing between components
- Code complexity analysis

3.2 Architecture Validation

Testing Method: White Box Testing

Testing Level: Integration Testing

- Microservices communication
- API endpoint functionality
- Database query optimization
- Caching mechanism effectiveness
- Service dependency validation

4. CHANGE-RELATED TESTING

4.1 Smoke Testing

Testing Level: System Testing

- Basic flight search functionality
- Main booking flow works
- Payment processing functions
- User login/authentication
- Core navigation elements

4.2 Sanity Testing

Testing Level: System Testing

- New feature integration validation
- Bug fix verification in specific areas
- Critical workflow functionality

- Data integrity after changes

4.3 Regression Testing

Testing Level: System Testing

- Existing search functionality unchanged
- Previous booking flows still work
- Payment integration stability
- User account management
- Performance benchmarks maintained

4.4 Retesting

Testing Level: Component/System Testing

- Execute exact failed test case
- Verify specific defect solution
- Confirm fix doesn't create new issues
- Check for root cause

5. ACCEPTANCE TESTING

I decided to move this section separately for convenience.

5.1 User Acceptance Testing (UAT)

Testing Method: Black Box Testing

- Complete booking journey validation
- Real payment transactions
- Mobile app usage scenarios
- Customer support integration
- Kiwi.com Guarantee features

5.2 Operational Acceptance Testing (OAT)

- System backup and restore
- Installation/upgrade procedures
- Disaster recovery protocols
- User management systems
- Data migration procedures
- Monitoring and alerting systems
- Performance under production load

5.3 Alpha Testing

- New feature validation
- Pre-release functionality testing
- Integration with partner airlines
- Beta program preparation

5.4 Beta Testing

- Real-world usage scenarios
- Mobile app beta versions
- New destination testing
- Feature feedback collection